

Exam

5/15/2026

 Legg til kommentar

5/15/2026

▼ Detaljer

Introduction

Group assignment: Choose a group under the Exam Groups category and add all members there. It does not have to be the same group as used for assignment deliveries.

This assignment will be graded and will determine your final grade in the course. As a graded exam, rules on cheating also applies here. See the [official rules on cheating](https://www.uia.no/student/eksamen/linkeblokksamling/under-eksamen/hva-regnes-som-fusk)  (<https://www.uia.no/student/eksamen/linkeblokksamling/under-eksamen/hva-regnes-som-fusk>) for details.

Make sure you get the required $\geq 80\%$ on assignments before the final assignment deadline. Any students with $<80\%$ on assignments will be withdrawn from the exam and will not be graded.

Delivery of this assignment is done in Inspira. Make sure you put together a group in Inspira before delivering. A group cannot have more than 4 members. After the group has been registered in Inspira, one group member on behalf of the group delivers the assignment, see section *Delivery* below for details.

Requirements

In this assignment you will create a smart clock. Your grade will be set based on which of the graded requirements you have completed. It is recommended to read over the requirements and planning so that you do the easiest ones without dependencies to other requirements first. Try to divide the work up in logical pieces among the members of the group.

You have a lot of freedom in how to implement these requirements. Look back on the course topics and use what you think makes sense. You can use threads, interrupts, events, mailboxes and so on.

Hint: The clock will have multiple "screens" for different information, you can use a button or two to switch the active screen.

Hint: Multiple requirements require automatic time, date, and location information. You can get this information from APIs like ipgeolocation.io, [↗ \(https://ipgeolocation.io/documentation/timezone-api.html\)](https://ipgeolocation.io/documentation/timezone-api.html) see *Using Client or Machine IP Address*.

Hint: To read JSON responses from APIs you can use the built in functionality in Zephyr. [See the documentation ↗ \(https://docs.zephyrproject.org/latest/services/serialization/json.html\)](https://docs.zephyrproject.org/latest/services/serialization/json.html).

Hint: There are multiple sites to get RSS feeds. Here's an example news feed from BBC: <http://feeds.bbc.co.uk/news/world/rss.xml> [↗ \(http://feeds.bbc.co.uk/news/world/rss.xml\)](http://feeds.bbc.co.uk/news/world/rss.xml).

Basic requirements

You get the grade F if any of these are missing.

- The clock must be implemented using Zephyr
- The clock must have a 16x2 display
- The clock must have a piezo speaker
- The clock must use at least 3 buttons. You choose how they are used, and you can add more buttons if you want

Graded requirements

Functional requirements

1. [10%] Use IP geolocation to retrieve Unix epoch time, time zone, daylight saving time, latitude, longitude, and city at start up (boot). Use the time information to set microcontroller Real Time Clock (RTC) and present the following information on screen:
 - a) Start with Unix epoch time for approx. 2 seconds e.g.:

```
Unix epoch time:  
1234567890
```

b) Then latitude and longitude for approx. 2 seconds e.g.:

```
Lat: 58.3405  
Lon: 8.5934
```

c) And finally city for approx. 2 seconds e.g.:

```
City:  
Grimstad
```

This information must be fetched using an API and information presented only once at startup. None of the above presented data, nor time zone and daylight saving time data, shall be hard coded in source code.

2. [10%] Add a screen that shows day of week, date, and month in addition to the time e.g.:

```
Fri 14 Apr 14:01
```

The information shall be presented on screen line 1 (top line) and would be the default date/time/alarm screen. When this screen is selected, its date/time information must be updated regularly.

3. [10%] Add a screen that lets the user set a recurring alarm time (e.g., 07:30) using buttons or any other input method. You can use UART RX from the PC terminal as input, but this will halve the maximum score to 5% on this requirement.

When the alarm is set, it is also enabled and shown on default screen line 2 (bottom line) e.g.:

```
Alarm      07:30
```

4. [5%] Sound an alarm at the set alarm time.

An active alarm shall, in addition to the sound, also be indicated on the default screen line 2 e.g.:

```
Alarm (A)  07:30
```

or e.g.:

Alarm active

To avoid this alarm to sound "forever", it shall be automatically muted after 10 minutes, and any active indications remove from default screen line 2.

5. [5%] Snooze alarm functionality.

Snoozes an active alarm for 5 minutes. This function shall not change the alarm set time, but it shall reset the automatically mute time. Meaning, when the alarm sound goes on after being snoozed for 5 minutes, it will again be automatically muted after 10 minutes. A snoozed alarm shall, in addition to sound off, also be indicated on default screen line 2 e.g.:

Alarm (S) 07:30

or e.g.:

Alarm snoozed

6. [5%] Mute alarm functionality.

Both a snoozed alarm and an active alarm may be muted. When the alarm is muted, the alarm is no longer active, but still enabled. Meaning, since this is a recurring alarm, it will go active at the set alarm time the following day.

A muted alarm shall have any snoozed or active indications removed and the set and enabled alarm re-appear on default screen line 2 e.g.:

Alarm 07:30

7. [5%] Enable/disable alarm functionality.

Enables/disables the alarm so it goes on/off at the configured time. When the alarm is disabled, it shall be removed from line 2 on the default screen. When the alarm is re-enabled, it shall re-appear on the default screen line 2 with the previously set alarm time e.g.:

Alarm 07:30

8. [10%] Add a screen that shows the current room temperature and humidity e.g.:

```
Temp: 21.4 C
Humidity: 69%
```

The information on this screen is on demand (e.g. navigated to with buttons) and shall be updated on a regular basis. Meaning if this screen is constantly shown, its information is refreshed and showing actual room temperature and humidity.

9. [10%] Add a screen that shows today's weather forecast.

This information must be fetched using an API (no faking it in code) e.g.:

```
Broken clouds
7 degrees
```

The information on this screen is on demand and shall be updated on a regular basis. Meaning when this screen is selected, the latest weather forecast is fetched and if this screen is constantly shown, its information is refreshed so that it shows latest actual weather forecast. Since weather forecast doesn't change that rapidly and to reduce network traffic, the refresh rate might be relatively low e.g., every 15 minutes.

10. [10%] Since IP geolocation is not always that accurate, add a screen allowing user to enter name of place/city or precise latitude/longitude coordinates to be used when fetching weather forecast. Any imaginable input method is valid, including UART RX. Using the UART for this will score 5%. If hard coded name of place/city or latitude/longitude coordinates are used, the score will be 0% on this requirement.
11. [10%] Add a screen that shows recent top news feeds. On screen line 1 (top line) the source of the news feeds is shown e.g., BBC or CNN. On screen line 2 (bottom line) allow the 3 top news feeds to scroll from right to left one after each other but separated by a few spaces or some symbol e.g., --- , -*- , # , * , or something else.

The information on this screen is on demand and shall be updated on a regular basis. Meaning when this screen is selected, the top news feeds are fetched and if this screen is constantly shown, its information is refreshed so that it shows latest actual top news.

This information must be fetched using an RSS feed (no faking it in code). RSS is a XML-based format, see https://www.w3schools.com/XML/xml_rss.asp  (https://www.w3schools.com/XML/xml_rss.asp).

You don't have to use an XML parser (they are a bit complicated), you can download the feed and jump from feed to feed by searching for tag `<item>`, then search for tag `<title>` to find the news headlines using standard string functions. Avoid feeds in Norwegian language since the DFRobot LCD driver does not handle Norwegian letters well.

Nonfunctional requirements

[10%] This requirement will assess the design and the structure of the source code. Criteria's this assessment will emphasize are:

- The design invites for group members to work on separate parts of the requirements e.g., the functionality is split up in separate source/header (*.cpp/*.h) files
- The design uses interrupts and callbacks, rather than extensive polling for events like e.g., button pressed
- The source code is readable with proper indentation, having a reasonable amount of comments and using explanatory names on variables and functions
- The source code is delivered as described in section Delivery below

Grading

- A 95% to 100%
- B 80% to <95%
- C 60% to <80%
- D 50% to <60%
- E 40% to <50%

- F 0% to <40%

Delivery

Delivery of this assignment is done in Inspera. The delivery is done in a separate set of groups, so make sure you put together a group in Inspera before delivering. A group cannot have more than 4 members. After the group has been registered in Inspera, one group member on behalf of the group delivers the assignment.

Deliver the following as a zip file (if any of these are missing you get the grade F):

- source
 - Directory with your source code.
 - Do NOT include the build and .idea directories. They are not needed and are very big.
- demo.mp4
 - Video clearly showcasing the device and what functionality is working.
 - You can record with your phone or with a webcam using [OBS Studio](https://obsproject.com/)  (<https://obsproject.com/>). Add a Video Capture Device (under Sources), select your webcam, then click "Start Recording".
 - You can talk over the video to explain if you want to, but it's not a requirement.
 - Length: 3 to 5 minutes
- checklist.pdf
 - List every requirement you have completed and shown in the video with their percentage
 - Show the sum of the percentages and the final grade